AN UNCONDITIONAL SIXTH-ENERGY SAVING FOR RIEMANN ZETA ZEROS AND A LOCAL PRIME-WINDOW EXCEPTIONAL-PAIR BOUND

ARGUS OPEN RESEARCH PROJECT

Abstract. We establish an unconditional power saving for the sixth additive energy $E_6(\sigma,T) = \#\{(\rho_1,\dots,\rho_6) \in Z(\sigma,T)^6 : |\gamma_1+\gamma_2+\gamma_3-\gamma_4-\gamma_5-\gamma_6| \le 1\}$ of nontrivial zeros of the Riemann zeta function in the neighborhood $|\sigma - 7/8| \le 1/1000$. By inserting a short mollified Dirichlet-polynomial zero detector at one zero in each sextuple and stratifying triple-sum multiplicity classes via weighted fifth- and sixth-ratio inequalities, we obtain a saving of $\delta = 1/200$ over the baseline density/fourth-energy interpolation envelope. Through the centered explicit formula transfer, this yields the unconditional local prime-window bound $\#\{(n,h) \in [N,2N)\times[H,2H) : \pi(n+h) > \pi(n)+\pi(h)\} \ll_\varepsilon N^{19/15-1/10000+\varepsilon}$ for dyadic lengths satisfying $|\log_N H - 5/12| \le 10^{-6}$.

1. Main Mathematical Results

Theorem 1.1 (Unconditional sixth-energy saving and local prime-window bound)**.** *Uniformly for $|\sigma - 7/8| \le 1/1000$ and $\varepsilon > 0$, the sixth additive energy of Riemann zeta zeros satisfies*

$$E_6(\sigma,T) \ \ll_\varepsilon \ T^{2d(\sigma)+e_4(\sigma)-\frac{1}{200}+\varepsilon}.$$

Consequently, for dyadic integers N, H with $|\log_N H - 5/12| \le 10^{-6}$, the number of Second Hardy–Littlewood exceptional pairs satisfies

$$V(N,H) \ = \ \#\big\{(n,h) \in [N,2N)\times[H,2H) : \pi(n+h) > \pi(n)+\pi(h)\big\} \ \ll_\varepsilon \ N^{\frac{19}{15}-\frac{1}{10000}+\varepsilon}.$$

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1518 in the domain of analytic number theory adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Unconditional power saving delta = 1/200 for the sixth additive energy E_6(sigma, T) of Riemann zeta zeros on |sigma - 7/8| <= 1/1000 and local exceptional-pair bound V(N,H) « N^(19/15 - 1/10000 + epsilon) near log_N H = 5/12.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Global resolution of the Second Hardy-Littlewood Conjecture pi(x+y) <= pi(x)+pi(y)

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 1518 (analytic number theory), archived in the accompanying repository package `results/1518/`.

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